NASA MICROGRAVITY COMBUSTION DATA — RAW CSV INPUT FIELD DOCUMENTATION

Purpose:

The raw NASA datasets contain different variables depending on the experiment. The variables below are taken from the CSV datasets provided for the project. The original NASA column names should be preserved in the raw data so that every value remains traceable to its source dataset. Variables that are unavailable in a particular experiment should be stored as NULL rather than being artificially estimated.

The raw datasets currently considered are:

- PSI-159 Normal Flames
- PSI-159 Inverse Flames
- PSI-25 BASS-II
- PSI-99 SAFFIRE-2
- PSI-107 SPICE
- PSI-100 SAFFIRE-3
- PSI-69 FLEX
- PSI-98 SAFFIRE-1

1. COMMON / EXPERIMENT IDENTIFICATION FIELDS

Source Dataset

Importance: Identifies the original NASA dataset from which the record came. This is essential for traceability and for understanding the physical conditions and measurement methods associated with the record.

Experiment

Importance: Identifies the particular NASA experiment or investigation, allowing records from different experiments to be distinguished.

Sample Number

Importance: Identifies the specific sample used during an experiment and allows multiple samples within the same experiment to be distinguished.

Sample #

Importance: Original sample identifier used by the dataset. It should be retained exactly for traceability to the NASA source.

Test #

Importance: Identifies the individual experimental test and allows the model output to be traced back to a specific experimental run.

FLEX Test #

Importance: Identifies the specific FLEX experiment test from which the record originated.

FLEX Identifier

Importance: Provides the original FLEX experiment identifier and allows individual experimental conditions to be traced back to the NASA dataset.

PI

Importance: Identifies the principal investigator associated with the experiment. It is metadata rather than a physical ML feature, but should be retained for provenance.

Test

Importance: Identifies the test or experimental run associated with the record.

Date

Importance: Records when the experiment was performed. It is useful for traceability and experimental history but should normally not be used as a physical prediction feature.

Test Date

Importance: Provides the experimental date in the original dataset format.

GMT

Importance: Records the experiment time in GMT and can be used to identify and trace an individual experimental run.

Test GMT

Importance: Provides the GMT timestamp associated with the test.

GMT date

Importance: Records the date associated with SPICE experimental observations.

GMT ignition

Importance: Records the ignition time of the SPICE experiment and helps identify the temporal sequence of the experiment.

GMT Flame Off

Importance: Records the time at which the SPICE flame was turned off or extinguished. It is experimental metadata and may also help define the experimental endpoint.

2. FUEL / MATERIAL FIELDS

Fuel

Importance: Identifies the fuel being burned. Different fuels have different chemical properties, evaporation behaviour, reaction chemistry and flame stability.

Fuel Sample Material

Importance: Identifies the solid material used in flame-spread experiments. Material composition strongly affects ignition, flame spread and extinction.

Material

Importance: Represents the material involved in a solid-fuel experiment and is important for distinguishing different combustion behaviours.

3. ATMOSPHERIC COMPOSITION FIELDS

Percent O2

Importance: Represents oxygen concentration in the surrounding atmosphere. Oxygen availability is one of the most important variables controlling ignition, flame spread, flame stability and extinction.

Percent O2 (Note 1)

Importance: Records the oxygen concentration under the specific BASS/SAFFIRE experimental definition. It should be retained because it is an original NASA field.

O initial ambient composition; mole fraction

Importance: Represents the initial oxygen composition of the ambient atmosphere in mole fraction. It describes the oxidizing environment before combustion.

N initial ambient composition; mole fraction

Importance: Represents the initial nitrogen concentration. Nitrogen acts primarily as a diluent and affects heat capacity, temperature and transport.

CO initial ambient composition; mole fraction

Importance: Records the initial carbon-monoxide fraction in the atmosphere. It is useful for reproducing the complete initial gas composition.

He initial ambient composition; mole fraction

Importance: Records the initial helium fraction. Helium can affect thermal conductivity and heat transport and therefore influence flame behaviour.

XO2 before test

Importance: Represents the oxygen mole fraction before the CFI-G experiment. It is a major predictor of whether combustion can remain sustained.

XO2 after test

Importance: Represents the oxygen concentration measured after the experiment. It is a post-experiment measurement and should generally be treated as an output/observation rather than an input.

Xf before test

Importance: Represents the fuel-related composition before the CFI-G experiment and describes the initial mixture condition.

Xf after test

Importance: Represents the fuel-related composition after the experiment. It is a post-experiment measurement and should generally not be used as a pre-test model input.

Calibrated initial O2 % by vol

Importance: Represents the calibrated oxygen concentration before the BASS-II experiment and provides the actual initial oxidizer condition.

Calibrated final O2 % by vol

Importance: Represents the oxygen concentration measured after the experiment. It is a post-experiment result and should not normally be used as a prediction input.

Initial CO2 % by vol

Importance: Represents the initial carbon-dioxide concentration. CO2 acts as a diluent and affects the thermal and chemical environment.

Final CO2 % by vol

Importance: Represents the carbon-dioxide concentration after combustion. It is an experimental outcome rather than a pre-test input.

Initial CO (ppm)

Importance: Represents the initial carbon-monoxide concentration in the atmosphere.

Final CO (ppm)

Importance: Represents the carbon-monoxide concentration after the experiment and should generally be treated as an experimental output.

4. PRESSURE FIELDS

Ambient pressure; mmHg

Importance: Ambient pressure affects chemical reaction rates, evaporation, diffusion and heat/mass transport and therefore influences flame stability and extinction.

approx p (bar)

Importance: Represents the experimental pressure in bar for the relevant CFI-G dataset. Pressure must be retained because it affects combustion behaviour and enables comparison between pressure conditions.

5. DROPLET COMBUSTION FIELDS

Droplet initial diameter; mm

Importance: Initial droplet size controls the surface-area-to-volume ratio, evaporation rate and amount of fuel available to sustain the flame. It is a key input for FLEX-type flame-extinction modelling.

Visible flame extinction diameter; mm

Importance: Represents the droplet diameter observed when the visible flame extinguishes. It is a combustion outcome/observation and should not be used as a pre-test input when predicting extinction.

Burning rate; mm

Importance: Represents the observed burning behaviour of the droplet. It is primarily a measured combustion outcome rather than an initial experimental input.

Burn time; s

Importance: Represents the duration of combustion. It can be used as a target/output for lifetime prediction but should not be used as an input when predicting the outcome before combustion.

6. FUEL AND OXIDIZER FLOW FIELDS

mHC (mg/s)

Importance: Represents the hydrocarbon mass-flow rate. It describes how much hydrocarbon fuel is supplied per second and directly affects the strength of combustion.

HC burnrate (mg/s)

Importance: Represents the hydrocarbon burning rate. It describes the amount of hydrocarbon being consumed and is useful for characterizing combustion intensity.

O2 flow rate (mg/s)

Importance: Represents the oxygen supply rate. Oxygen availability at the reaction zone strongly affects flame sustainability and extinction.

Fuel Rate (sccm)

Importance: Represents volumetric fuel-flow rate in standard cubic centimetres per minute. It determines the amount of fuel supplied and affects mixture conditions.

Fuel Rate (mg/s)

Importance: Represents the mass fuel-flow rate. It is useful for comparing experiments because it directly represents the mass of fuel supplied per unit time.

HC Fuel Rate (mg/s)

Importance: Represents the hydrocarbon fuel mass-flow rate and is directly related to the amount of carbon-containing fuel available for combustion and soot formation.

Air Rate (cm/s)

Importance: Represents the air-flow condition in the SPICE experiment. Airflow controls fuel-air mixing and oxygen transport and therefore affects flame and soot behaviour.

Air Flow (cm/s)

Importance: Represents the forced airflow velocity around the sample. Airflow changes heat transfer, oxygen transport and flame propagation.

Air Flow

Importance: Original SAFFIRE airflow field. It should be retained exactly because it represents the airflow condition reported by that dataset.

7. FLOW CONFIGURATION FIELDS

Flow Direction

Importance: Indicates the direction of airflow relative to flame propagation. Concurrent and opposed flows can produce substantially different flame-spread behaviour.

Direction

Importance: Original SAFFIRE flow-direction field. It should be retained because it identifies the experimental flow configuration.

Flow restrictor

Importance: Represents the flow-control configuration used in BASS-II. It helps describe the actual airflow conditions experienced by the sample.

Fan display

Importance: Records the fan setting used in the BASS-II experiment and can be used to reconstruct the experimental airflow condition.

Air display

Importance: Records the air-system setting associated with the BASS-II experiment and provides additional information about the imposed flow condition.

8. SOLID SAMPLE GEOMETRY FIELDS

Samle Thickness

Importance: Represents the thickness of the solid sample. Thickness affects thermal mass, heat conduction and the amount of material available for combustion.

Sample Thickness (cm)

Importance: Represents sample thickness in centimetres. It is an important geometric and thermal parameter for flame-spread prediction.

Sample Thickness

Importance: Original SAFFIRE sample-thickness field. It should be retained because different SAFFIRE datasets use different field formats.

Sample Length (cm)

Importance: Represents the physical length of the sample. It defines the available distance for flame propagation and contributes to the sample geometry.

Sample Length

Importance: Original sample-length field used by the relevant SAFFIRE dataset.

Sample Width (cm)

Importance: Represents sample width in centimetres. It defines the exposed sample geometry and can influence flame growth and heat transfer.

Sample Width

Importance: Original sample-width field used by the relevant SAFFIRE dataset.

9. IGNITION FIELDS

Ignition Power (W)

Importance: Represents the power supplied by the ignition source. Ignition energy can affect whether the material ignites and the initial strength of the flame.

Ignition Time (s)

Importance: Represents how long the ignition source is applied. It determines the amount of energy transferred to the material before flame development.

10. COMBUSTION REGIME / CHEMISTRY FIELDS

Cool flame?

Importance: Indicates whether the experiment involves a cool-flame regime. Cool flames can have substantially different chemistry and temperature behaviour from conventional hot flames.

Hot ignition

Importance: Indicates whether hot ignition occurred. It is useful for analysing ignition behaviour but should only be used as an input if its value is known before the predicted event.

TFP

Importance: Original CFI-G field. Its exact physical meaning should be retained from the NASA documentation and verified before using it as a numerical ML feature.

Tad (K)

Importance: Represents adiabatic flame temperature. It describes the theoretical thermal state of the mixture and can be useful as a physics-derived feature, but it may be derived from other input conditions.

Zst

Importance: Represents a stoichiometric mixture-related parameter. It helps describe the mixture condition relative to stoichiometric combustion.

11. SOOT / SMOKE-SPECIFIC FIELDS

Nozzle (mm)

Importance: Nozzle diameter determines the geometry of the fuel jet and influences fuel-air mixing, flame structure and soot formation.

Fuel Rate (sccm)

Importance: Fuel supply rate affects mixture richness, flame size and the quantity of carbon-containing fuel available for soot production.

Fuel Rate (mg/s)

Importance: Mass fuel-flow rate provides a direct measure of fuel supplied per unit time and is useful for comparing soot-producing conditions.

HC Fuel Rate (mg/s)

Importance: Hydrocarbon mass-flow rate directly describes the amount of hydrocarbon fuel entering the flame and therefore affects soot formation potential.

Air Rate (cm/s)

Importance: Airflow affects fuel-air mixing and oxygen availability. These strongly influence whether soot forms, grows or is oxidized.

12. GRAVITY / EXPERIMENTAL ENVIRONMENT

Gravity Level

Importance: Gravity changes buoyancy-driven flow and therefore affects heat and mass transport around the flame. Microgravity is especially important because diffusion and forced flow become more dominant.

Note:

Gravity Level is part of the normalized project-level schema described in the project documentation, but it should not be artificially inserted into a raw CSV if the original NASA CSV does not explicitly contain that field. Instead, it should be added during normalization using the known experiment configuration.

13. FIELDS THAT SHOULD GENERALLY BE TREATED AS OUTPUTS / OBSERVATIONS

Visible flame extinction diameter; mm

Importance: This is an observed extinction result rather than a pre-experiment condition.

Burning rate; mm

Importance: This represents measured combustion behaviour and can be used as a prediction target or analysis variable.

Burn time; s

Importance: This is the observed duration of combustion and can be used as a target for flame-lifetime prediction.

burn time (sec)

Importance: Same physical concept as Burn time; s in the relevant dataset and should be treated as an observed combustion duration.

XO2 after test

Importance: Represents the measured oxygen condition after combustion and is therefore a post-experiment variable.

Xf after test

Importance: Represents the measured post-experiment fuel-related composition.

Calibrated final O2 % by vol

Importance: Represents oxygen concentration after the experiment and is therefore a result of the combustion process.

Final CO2 % by vol

Importance: Represents post-combustion CO2 concentration and should be treated as an observed outcome.

Final CO (ppm)

Importance: Represents post-combustion CO concentration and should be treated as an observed outcome.

?-g Spread Length

Importance: Represents observed flame-spread distance and should be used as a target/output for flame-spread prediction.

1-g Spread Rate

Importance: Represents observed flame-spread rate and is one of the main possible prediction targets for the flame-spread model.

?-g Burn Length

Importance: Represents observed burning length under the relevant gravity condition and is an experimental outcome.

1-g Burn Length

Importance: Represents observed burning length under normal gravity and is an experimental outcome.

Burn Time

Importance: Represents observed combustion duration and can be used as an output for flame-lifetime prediction.

Test end

Importance: Indicates the experimental endpoint and can help define whether the flame continued, extinguished or the test was manually terminated.

14. METADATA / TRACEABILITY FIELDS

GMT date

Importance: Used to identify the date of the experiment and maintain traceability.

GMT ignition

Importance: Used to identify the ignition event in the SPICE experiment.

GMT Flame Off

Importance: Used to identify the flame-off event and can help calculate experimental flame duration.

Color

Importance: Descriptive metadata from the original dataset. It should be retained for traceability but is not normally a physical ML feature.

Comments

Importance: Preserves experimental notes and observations that may contain important context not represented by numerical variables.

Camera 1

Importance: Identifies the camera/video source associated with the experiment. It is useful for future computer-vision analysis and traceability.

Camera 2

Importance: Identifies a second camera/video source and can support future flame-image or flame-video analysis.

15. RECOMMENDED NORMALIZED INPUT SCHEMA FOR THE THREE ML MODELS

The raw CSV must preserve the original NASA column names. After ingestion, a separate normalized dataset should map compatible variables into common project-level names.

FLAME EXTINCTION — NORMALIZED INPUTS

Fuel

Importance: Identifies the combustion fuel and captures differences in chemical and physical behaviour.

Oxygen Concentration

Importance: Describes oxidizer availability and is one of the strongest factors controlling flame sustainability and extinction.

Pressure

Importance: Affects reaction rates, evaporation, diffusion and heat transfer.

Fuel Concentration / Mixture Composition

Importance: Describes the fuel-air mixture and determines how close the system is to conditions that can sustain combustion.

Fuel Flow Rate

Importance: Determines the amount of fuel supplied to the reaction zone.

Oxygen Flow Rate

Importance: Determines the rate at which oxidizer is supplied.

Droplet Initial Diameter

Importance: Controls evaporation, surface-area-to-volume ratio and available fuel mass in droplet combustion.

Gravity Level

Importance: Determines the importance of buoyancy-driven transport versus diffusion and forced convection.

FLAME SPREAD — NORMALIZED INPUTS

Material

Importance: Different solid materials have different thermal and combustion properties.

Oxygen Concentration

Importance: Controls the ability of the solid material to sustain combustion and affects flame-spread rate.

Air Flow

Importance: Controls oxygen transport, heat transfer and flame propagation.

Flow Direction

Importance: Distinguishes concurrent and opposed-flow configurations, which can produce different spread rates.

Pressure

Importance: Affects combustion and transport processes.

Sample Thickness

Importance: Affects thermal mass, heat conduction and available fuel.

Sample Length

Importance: Defines the physical geometry available for flame propagation.

Sample Width

Importance: Defines the exposed sample geometry.

Ignition Power

Importance: Determines the energy initially supplied to ignite the material.

Ignition Time

Importance: Determines how long the material is exposed to the ignition source.

Gravity Level

Importance: Determines the role of buoyancy in flame growth and spread.

SOOT / SMOKE — NORMALIZED INPUTS

Fuel

Importance: Fuel chemistry strongly influences soot formation and oxidation.

Oxygen Concentration

Importance: Controls oxidation and affects whether soot is produced or consumed.

Fuel Flow Rate

Importance: Determines the amount of carbon-containing fuel entering the flame.

Hydrocarbon Fuel Rate

Importance: Represents the actual hydrocarbon supply and is directly relevant to soot formation.

Air Flow

Importance: Controls fuel-air mixing and oxygen transport.

Nozzle Diameter

Importance: Determines the fuel-jet geometry and mixing behaviour.

Pressure

Importance: Influences combustion chemistry, transport and soot formation.

Temperature

Importance: Influences reaction rates and soot formation/oxidation.

Geometry

Importance: Defines the flame configuration and affects mixing and soot behaviour.

Gravity Level

Importance: Changes buoyancy-driven transport and therefore influences flame structure and smoke/soot transport.

16. IMPORTANT DATA-DESIGN RULE

Do not force all NASA datasets into one identical feature set.

The RAW dataset should preserve the original NASA column names.

The NORMALIZED dataset should contain common project-level variables.

Variables unavailable in a particular experiment should be stored as NULL.

Variables measured after combustion should not be used as pre-experiment ML inputs.

For example:

Droplet initial diameter → INPUT

Visible flame extinction diameter → OUTPUT

Initial O2 → INPUT

Final O2 → OUTPUT

Sample thickness → INPUT

Flame spread rate → OUTPUT

Fuel rate → INPUT

Burn time → OUTPUT

This separation is essential to prevent data leakage and to ensure that the ML model only uses information that would actually be available when making a prediction.

The three task-specific models should therefore use different subsets of the normalized variables:

Flame Extinction:

Fuel + Oxygen Concentration + Pressure + Mixture Composition + Fuel/Oxygen Flow + Droplet Size + Gravity

Flame Spread:

Material + Oxygen Concentration + Air Flow + Flow Direction + Pressure + Sample Geometry + Ignition Conditions + Gravity

Soot / Smoke:

Fuel + Oxygen Concentration + Fuel Flow + Hydrocarbon Fuel Rate + Air Flow + Nozzle Geometry + Pressure + Temperature + Flame Geometry + Gravity

INPUT FIELD → NASA EXPERIMENT → PROJECT CATEGORY

CATEGORY 1: FLAME EXTINCTION

Fuel → PSI-69 FLEX → Flame Extinction

Fuel → PSI-159 Normal Flames → Flame Extinction

Fuel → PSI-159 Inverse Flames → Flame Extinction

Ambient pressure; mmHg → PSI-69 FLEX → Flame Extinction

O initial ambient composition; mole fraction → PSI-69 FLEX → Flame Extinction

N initial ambient composition; mole fraction → PSI-69 FLEX → Flame Extinction

CO initial ambient composition; mole fraction → PSI-69 FLEX → Flame Extinction

He initial ambient composition; mole fraction → PSI-69 FLEX → Flame Extinction

Droplet initial diameter; mm → PSI-69 FLEX → Flame Extinction

XO2 before test → PSI-159 Normal Flames → Flame Extinction

XO2 before test → PSI-159 Inverse Flames → Flame Extinction

approx p (bar) → PSI-159 Normal Flames → Flame Extinction

approx p (bar) → PSI-159 Inverse Flames → Flame Extinction

Xf before test → PSI-159 Normal Flames → Flame Extinction

Xf before test → PSI-159 Inverse Flames → Flame Extinction

mHC (mg/s) → PSI-159 Normal Flames → Flame Extinction

mHC (mg/s) → PSI-159 Inverse Flames → Flame Extinction

HC burnrate (mg/s) → PSI-159 Normal Flames → Flame Extinction

HC burnrate (mg/s) → PSI-159 Inverse Flames → Flame Extinction

O2 flow rate (mg/s) → PSI-159 Normal Flames → Flame Extinction

O2 flow rate (mg/s) → PSI-159 Inverse Flames → Flame Extinction

Cool flame? → PSI-159 Normal Flames → Flame Extinction

Cool flame? → PSI-159 Inverse Flames → Flame Extinction

Hot ignition → PSI-159 Normal Flames → Flame Extinction

Hot ignition → PSI-159 Inverse Flames → Flame Extinction

TFP → PSI-159 Normal Flames → Flame Extinction

TFP → PSI-159 Inverse Flames → Flame Extinction

Tad (K) → PSI-159 Normal Flames → Flame Extinction

Tad (K) → PSI-159 Inverse Flames → Flame Extinction

Zst → PSI-159 Normal Flames → Flame Extinction

Zst → PSI-159 Inverse Flames → Flame Extinction

CATEGORY 2: FLAME SPREAD

Fuel Sample Material → PSI-25 BASS-II → Flame Spread

Material → PSI-98 SAFFIRE-1 → Flame Spread

Material → PSI-99 SAFFIRE-2 → Flame Spread

Material → PSI-100 SAFFIRE-3 → Flame Spread

Samle Thickness → PSI-25 BASS-II → Flame Spread

Sample Thickness (cm) → PSI-98 SAFFIRE-1 → Flame Spread

Sample Thickness → PSI-100 SAFFIRE-3 → Flame Spread

Sample Length (cm) → PSI-98 SAFFIRE-1 → Flame Spread

Sample Length → PSI-100 SAFFIRE-3 → Flame Spread

Sample Width (cm) → PSI-98 SAFFIRE-1 → Flame Spread

Sample Width → PSI-100 SAFFIRE-3 → Flame Spread

Air Flow (cm/s) → PSI-25 BASS-II → Flame Spread

Air Flow → PSI-98 SAFFIRE-1 → Flame Spread

Air Flow → PSI-99 SAFFIRE-2 → Flame Spread

Air Flow → PSI-100 SAFFIRE-3 → Flame Spread

Air display → PSI-25 BASS-II → Flame Spread

Fan display → PSI-25 BASS-II → Flame Spread

Flow restrictor → PSI-25 BASS-II → Flame Spread

Percent O2 (Note 1) → PSI-25 BASS-II → Flame Spread

Percent O2 → PSI-98 SAFFIRE-1 → Flame Spread

Percent O2 → PSI-99 SAFFIRE-2 → Flame Spread

Percent O2 → PSI-100 SAFFIRE-3 → Flame Spread

Ignition Power (W) → PSI-98 SAFFIRE-1 → Flame Spread

Ignition Power (W) → PSI-99 SAFFIRE-2 → Flame Spread

Ignition Power (W) → PSI-100 SAFFIRE-3 → Flame Spread

Ignition Time (s) → PSI-98 SAFFIRE-1 → Flame Spread

Ignition Time (s) → PSI-99 SAFFIRE-2 → Flame Spread

Ignition Time (s) → PSI-100 SAFFIRE-3 → Flame Spread

Flow Direction → PSI-98 SAFFIRE-1 → Flame Spread

Flow Direction → PSI-99 SAFFIRE-2 → Flame Spread

Direction → PSI-100 SAFFIRE-3 → Flame Spread

Calibrated initial O2 % by vol → PSI-25 BASS-II → Flame Spread

Initial CO2 % by vol → PSI-25 BASS-II → Flame Spread

Initial CO (ppm) → PSI-25 BASS-II → Flame Spread

CATEGORY 3: SOOT / SMOKE

Fuel → PSI-107 SPICE → Soot / Smoke

Nozzle (mm) → PSI-107 SPICE → Soot / Smoke

Fuel Rate (sccm) → PSI-107 SPICE → Soot / Smoke

Fuel Rate (mg/s) → PSI-107 SPICE → Soot / Smoke

HC Fuel Rate (mg/s) → PSI-107 SPICE → Soot / Smoke

Air Rate (cm/s) → PSI-107 SPICE → Soot / Smoke

NORMALIZED PROJECT-LEVEL INPUTS AND THEIR SOURCE DATASETS

FLAME EXTINCTION

Fuel

→ PSI-69 FLEX

→ PSI-159 Normal Flames

→ PSI-159 Inverse Flames

Oxygen Concentration

→ PSI-69 FLEX

→ PSI-159 Normal Flames

→ PSI-159 Inverse Flames

Pressure

→ PSI-69 FLEX

→ PSI-159 Normal Flames

→ PSI-159 Inverse Flames

Fuel Concentration / Mixture Composition

→ PSI-159 Normal Flames

→ PSI-159 Inverse Flames

Nitrogen Concentration

→ PSI-69 FLEX

CO Concentration

→ PSI-69 FLEX

Helium Concentration

→ PSI-69 FLEX

Fuel Flow Rate

→ PSI-159 Normal Flames

→ PSI-159 Inverse Flames

Oxygen Flow Rate

→ PSI-159 Normal Flames

→ PSI-159 Inverse Flames

Droplet Initial Diameter

→ PSI-69 FLEX

Gravity Level

→ Experiment-level metadata associated with the relevant NASA experiment; not an original common column in all raw CSVs.

FLAME SPREAD

Material

→ PSI-25 BASS-II

→ PSI-98 SAFFIRE-1

→ PSI-99 SAFFIRE-2

→ PSI-100 SAFFIRE-3

Oxygen Concentration

→ PSI-25 BASS-II

→ PSI-98 SAFFIRE-1

→ PSI-99 SAFFIRE-2

→ PSI-100 SAFFIRE-3

Air Flow

→ PSI-25 BASS-II

→ PSI-98 SAFFIRE-1

→ PSI-99 SAFFIRE-2

→ PSI-100 SAFFIRE-3

Flow Direction

→ PSI-98 SAFFIRE-1

→ PSI-99 SAFFIRE-2

→ PSI-100 SAFFIRE-3

Pressure

→ No directly common pressure field identified in the flame-spread CSV input fields above; retain only where the original dataset provides it.

Sample Thickness

→ PSI-25 BASS-II

→ PSI-98 SAFFIRE-1

→ PSI-100 SAFFIRE-3

Sample Length

→ PSI-98 SAFFIRE-1

→ PSI-100 SAFFIRE-3

Sample Width

→ PSI-98 SAFFIRE-1

→ PSI-100 SAFFIRE-3

Ignition Power

→ PSI-98 SAFFIRE-1

→ PSI-99 SAFFIRE-2

→ PSI-100 SAFFIRE-3

Ignition Time

→ PSI-98 SAFFIRE-1

→ PSI-99 SAFFIRE-2

→ PSI-100 SAFFIRE-3

Gravity Level

→ Experiment-level metadata associated with the relevant SAFFIRE/BASS experiment.

SOOT / SMOKE

Fuel

→ PSI-107 SPICE

Oxygen Concentration

→ Not directly represented by the SPICE input fields listed above; should be added only when available from the corresponding NASA source data.

Fuel Flow Rate

→ PSI-107 SPICE

Hydrocarbon Fuel Rate

→ PSI-107 SPICE

Air Flow

→ PSI-107 SPICE

Nozzle Diameter

→ PSI-107 SPICE

Pressure

→ Not directly represented by the SPICE input fields listed above; should be added only when available from the corresponding NASA source data.

Temperature

→ Not directly represented by the SPICE input fields listed above.

Geometry

→ Not directly represented by the SPICE input fields listed above; Nozzle (mm) is the available geometric parameter.

Gravity Level

→ Experiment-level metadata associated with the SPICE experiment.

IMPORTANT RAW-DATA RULE

The following mapping should be maintained:

PSI-69 FLEX

→ Flame Extinction

PSI-159 Normal Flames

→ Flame Extinction

PSI-159 Inverse Flames

→ Flame Extinction

PSI-25 BASS-II

→ Flame Spread

→ Can also provide extinction-related observations, but its primary project category is Flame Spread.

PSI-98 SAFFIRE-1

→ Flame Spread

PSI-99 SAFFIRE-2

→ Flame Spread

PSI-100 SAFFIRE-3

→ Flame Spread

PSI-107 SPICE

→ Soot / Smoke

PSI-102 SAME

→ Soot / Smoke

→ Additional dataset to be incorporated separately when its experimental table is obtained.

IMPORTANT:

A field should only be used as an ML INPUT if it is known before the prediction is made.

For example:

Droplet initial diameter; mm

→ INPUT

Visible flame extinction diameter; mm

→ OUTPUT

XO2 before test

→ INPUT

XO2 after test

→ OUTPUT

Calibrated initial O2 % by vol

→ INPUT

Calibrated final O2 % by vol

→ OUTPUT

Sample Thickness

→ INPUT

1-g Spread Rate

→ OUTPUT

Fuel Rate (mg/s)

→ INPUT

Burn Time

→ OUTPUT

This separation prevents data leakage and ensures that the model predicts the combustion behaviour using only information that would actually be available before the experiment outcome is known.